



Argument Passing

OPERATING SYSTEM AND SYSTEM PROGRAMMING

V Vishwa Teja

2520090033

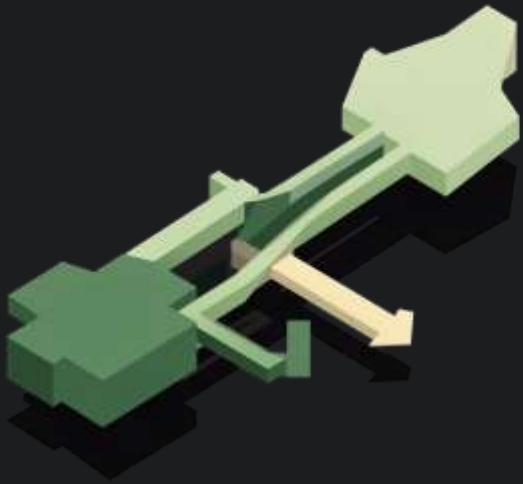
B Sree Nikhil Reddy

2520090058

Nalla Rohith

2520090191

Abstract



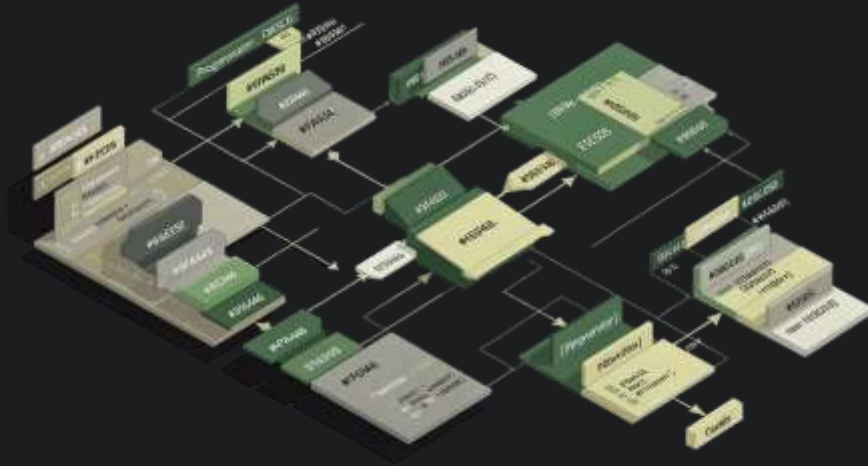
Argument passing is an important concept in programming that allows information to be transferred from one function to another. When a function is called, we can provide it with certain values, known as arguments, which the function uses to perform a specific task. This makes programs easier to understand, organize, and reuse. There are mainly two common ways of passing arguments: **pass by value** and **pass by reference**. In pass by value, a copy of the original value is given to the function. Any changes made to that value inside the function will not affect the original variable. On the other hand, pass by reference provides the function with access to the original variable or its reference. Therefore, changes made inside the function can also affect the original data.

Argument passing is useful because it allows programmers to create flexible functions that can work with different inputs instead of writing separate code for every situation. It also helps reduce repeated code and makes programs more modular and manageable. Different programming languages handle argument passing in different ways. For example, C commonly uses pointers to achieve reference-like behavior, while Java passes primitive values by value and object references by value. Python also passes object references by assignment.

Introduction



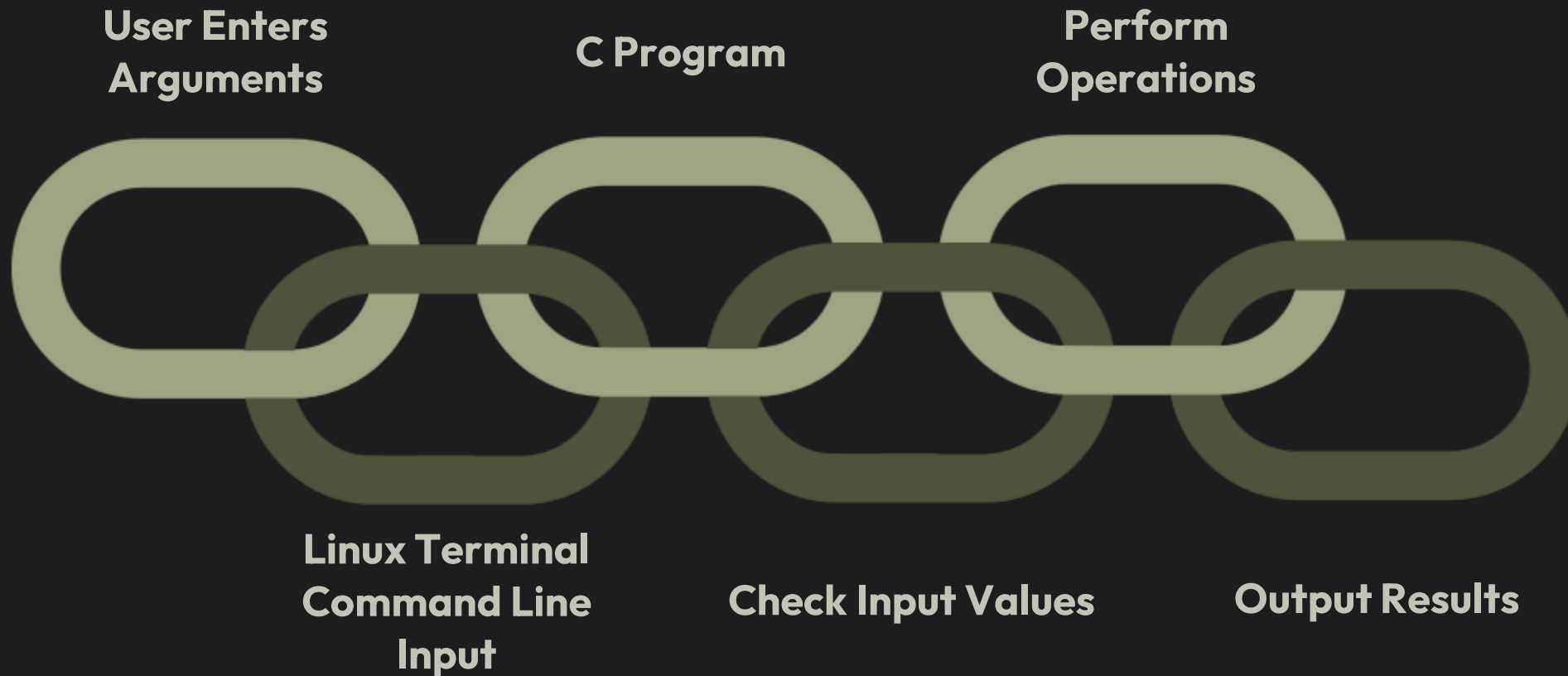
- **Argument Passing** is an important concept in Linux and C programming that allows users to provide information to a program directly through the command line during execution. Instead of entering data after the program starts, users can pass the required values along with the program name. This makes programs more flexible, convenient, and useful for different types of tasks.
- This project demonstrates how command-line arguments are received and processed using the `argc` and `argv` parameters of the C `main()` function. The program can identify the number of arguments, display their values, validate the input, and perform operations based on the arguments provided by the user.
- For example, users can execute a program by providing numbers, file names, or other values as arguments. The program then processes these values according to its functionality. This concept is commonly used in Linux commands, shell scripts, compilers, and system utilities.
- The project provides practical knowledge of command-line programming, input handling, string processing, and program execution. It also helps in understanding how the Linux shell communicates with applications. Overall, this project offers a simple and practical way to learn the fundamentals of argument passing and its role in Operating Systems and Linux programming.



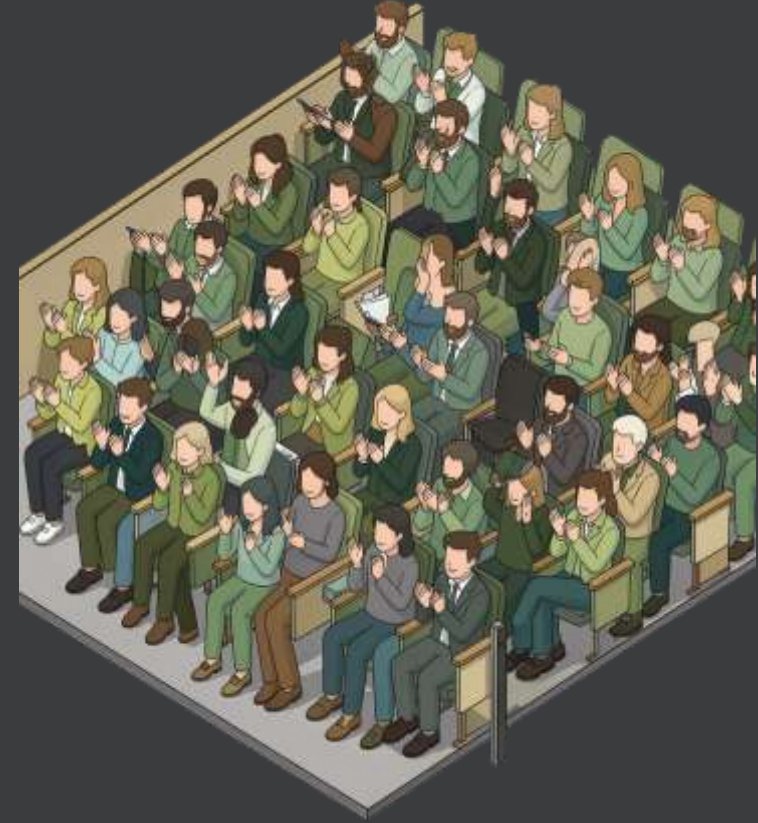
Hardware & Software Requirements

Requirement Type	Specification
Processor	Intel Core i3 or equivalent
Compiler	GCC
Storage	500 MB free space
Operating System	Windows / Linux / Ubuntu
Programming Language	C/C++
IDE	VS Code / Code::Blocks / Dev-C++
Terminal	Linux Terminal

System Architecture



The **Argument Passing System** follows a simple architecture where the user provides arguments through the Linux terminal, and the C program receives, validates, and processes them.



Thank You.